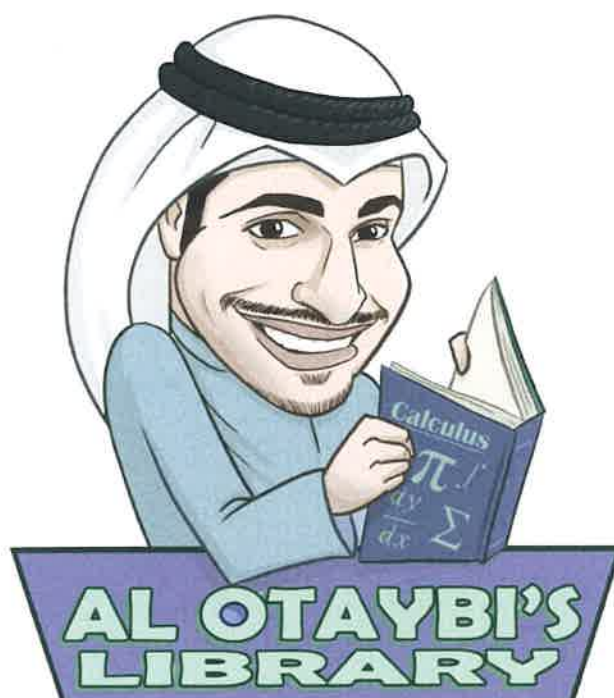


Kuwait University

2021/2022

Fall course



Calculus A

First

Sections : 2.3 (part b) – 2.2

2.3 Calculating Limits Using the Limit Laws

2.2 The Limit of a Function

(6)



Special case



إذا عندي مطلق يجب أن يتم إعادة تعريفه قبل حله

هل الممت تسعى لصفره ??

لا

يتم التعرف على اشارته

نعم

يتم دراسته من الاتجاهين

$$\lim_{x \rightarrow 5} |x - 3|$$

$$\lim_{x \rightarrow 5} + (x - 3) = 5 - 3 = 2$$

$$\lim_{x \rightarrow 2} |x - 3|$$

$$\lim_{x \rightarrow 2} - (x - 3) = - (2 - 3) = -(-1) = 1$$



بالمثل مع الجذر الزوجي

Even Root



$$\lim_{x \rightarrow 0} \frac{|x^2 - 1|}{x^2 - 1}$$

$$= \lim_{x \rightarrow 0} \frac{-(x^2 - 1)}{x^2 - 1}$$

$$= \lim_{x \rightarrow 0} -1 = -1$$

$$\lim_{x \rightarrow 4} |x - 4|$$

$$\lim_{x \rightarrow 4^+} (x - 4) = 0 \quad | \quad \lim_{x \rightarrow 4^-} -(x - 4) = 0$$

$$\therefore \lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^-} f(x) = 0$$

$$\therefore \lim_{x \rightarrow 4} f(x) = 0$$

$$\lim_{x \rightarrow 4} |4 - x|$$

$$\lim_{x \rightarrow 4^+} -(4 - x) = 0 \quad | \quad \lim_{x \rightarrow 4^-} (4 - x) = 0$$

$$\therefore \lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^-} f(x) = 0$$

$$\therefore \lim_{x \rightarrow 4} f(x) = 0$$

$$\lim_{x \rightarrow 1} \frac{|x-1|}{x^2+x-2}$$

$$\lim_{x \rightarrow 1^+} \frac{+(x-1)}{x^2+x-2} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 1^+} \frac{\cancel{(x-1)}}{\cancel{(x-1)}(x+2)}$$

$$\lim_{x \rightarrow 1^+} \frac{1}{x+2} = \frac{1}{3}$$

$$\lim_{x \rightarrow 1^-} \frac{-(x-1)}{x^2+x-2} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 1^-} \frac{-\cancel{(x-1)}}{\cancel{(x-1)}(x+2)}$$

$$\lim_{x \rightarrow 1^-} \frac{-1}{x+2} = -\frac{1}{3}$$

$$\therefore \lim_{x \rightarrow 1^+} f(x) \neq \lim_{x \rightarrow 1^-} f(x)$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1} f(x) \text{ DNE}$$

$$\lim_{x \rightarrow 2} \frac{3x^2-12}{|2-x|}$$

$$\lim_{x \rightarrow 2^+} \frac{3x^2-12}{-(2-x)} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 2^+} \frac{3(x^2-4)}{-(2-x)}$$

$$\lim_{x \rightarrow 2^+} \frac{3(x-2)(x+2)}{-(2-x)}$$

$$\lim_{x \rightarrow 2^+} \frac{3\cancel{(x-2)}(x+2)}{\cancel{-(x-2)}} \quad \text{لأنه } \ominus \text{ عامل مشترك}$$

$$= \frac{3(4)}{1} = 12$$

$$\lim_{x \rightarrow 2^-} \frac{3x^2-12}{+(2-x)}$$

$$\lim_{x \rightarrow 2^-} \frac{3(x^2-4)}{(2-x)}$$

$$\lim_{x \rightarrow 2^-} \frac{3(x-2)(x+2)}{(2-x)}$$

$$\lim_{x \rightarrow 2^-} \frac{3\cancel{(x-2)}(x+2)}{\cancel{-(x-2)}}$$

$$= \frac{3(4)}{-1} = -12$$

$$\therefore \lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$$

$$\Rightarrow \therefore \lim_{x \rightarrow 2} f(x) \text{ DNE}$$



Text book questions



$$\text{Let } g(x) = \frac{x^2 + x - 6}{|x - 2|}$$

a) Find $\lim_{x \rightarrow 2^+} g(x)$

b) Find $\lim_{x \rightarrow 2^-} g(x)$

c) Does $\lim_{x \rightarrow 2} g(x)$ exist?

$$\begin{aligned} \text{a) } \lim_{x \rightarrow 2^+} \frac{x^2 + x - 6}{|x - 2|} &= \frac{0}{0} \\ \lim_{x \rightarrow 2^+} \frac{(x-2)(x+3)}{(x-2)} &= 5 \end{aligned}$$

$$\begin{aligned} \text{b) } \lim_{x \rightarrow 2^-} \frac{x^2 + x - 6}{|x - 2|} &= \frac{0}{0} \\ \lim_{x \rightarrow 2^-} \frac{(x-2)(x+3)}{-(x-2)} &= -5 \end{aligned}$$

$$\text{c) } \because \lim_{x \rightarrow 2^+} g(x) \neq \lim_{x \rightarrow 2^-} g(x)$$

$$\therefore \lim_{x \rightarrow 2} g(x) \text{ DNE}$$



Text book questions



$$\lim_{x \rightarrow -2} \frac{2 - |x|}{2 + x}$$

$$\lim_{x \rightarrow -2} \frac{2 - (-x)}{2 + x}$$

$$\lim_{x \rightarrow -2} \frac{\cancel{2+x}}{\cancel{2+x}}$$

$$\lim_{x \rightarrow -2} (1) = 1$$

$$\lim_{x \rightarrow -6} \frac{2x + 12}{|x + 6|}$$

$$\lim_{x \rightarrow -6^+} \frac{2x + 12}{(x + 6)} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow -6^+} \frac{\cancel{2(x+6)}}{\cancel{x+6}}$$

$$= 2$$

$$\lim_{x \rightarrow -6^-} \frac{2x + 12}{-(x + 6)}$$

$$\lim_{x \rightarrow -6^-} \frac{\cancel{2(x+6)}}{-(\cancel{x+6})}$$

$$= -2$$

$$\therefore \lim_{x \rightarrow -6^+} \frac{2x + 12}{|x + 6|} \neq \lim_{x \rightarrow -6^-} \frac{2x + 12}{|x + 6|}$$

$$\therefore \lim_{x \rightarrow -6} \frac{2x + 12}{|x + 6|} \text{ DNE}$$



Notes



$$\sqrt{\frac{a+b}{c}}$$

$$= \frac{\sqrt{a+b}}{\sqrt{c}}$$

$$= \frac{\sqrt{a} + \sqrt{b}}{\sqrt{c}}$$



الأس يتوزع
على الضرب
والقسمة فقط



الأس لا يتوزع
على الجمع
والطرح



$$\sqrt{x} \text{ زوجي} = |x|$$

Examples

$$\sqrt{x^2} = |x|$$

$$\sqrt[3]{x^3} = x$$

$$\sqrt[4]{x^4} = |x|$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} x^2 \sqrt{1 + \frac{1}{x^2}}$$

$$0 \cdot \sqrt{1 + \frac{1}{0}}$$

$$0 \cdot \sqrt{1 + \infty}$$

$$0 \cdot \sqrt{\infty}$$

$$0 \cdot \infty$$

$$= \lim_{x \rightarrow 0} x^2 \sqrt{\frac{x^2}{x^2} + \frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0} x^2 \sqrt{\frac{x^2 + 1}{x^2}}$$

$$= \lim_{x \rightarrow 0} x^2 \frac{\sqrt{x^2 + 1}}{\sqrt{x^2}}$$

$$= \lim_{x \rightarrow 0} x^2 \frac{\sqrt{x^2 + 1}}{|x|}$$

$$\lim_{x \rightarrow 0^+} x^2 \frac{\sqrt{x^2 + 1}}{(x)}$$

$$\lim_{x \rightarrow 0^-} x^2 \frac{\sqrt{x^2 + 1}}{-(x)}$$

$$\lim_{x \rightarrow 0^+} x \sqrt{x^2 + 1} \quad \lim_{x \rightarrow 0^-} x \sqrt{x^2 + 1}$$

$$= 0 \sqrt{1} = 0$$

$$= \frac{0 \sqrt{1}}{-1} = 0$$

$$\therefore \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = 0$$

$$\therefore \lim_{x \rightarrow 0} f(x) = 0$$



2.2 The Limit of a Function



Evaluate the limit if it exist

$$\lim_{x \rightarrow 4} \frac{2}{x-4}$$

$$\frac{2}{\infty} = \infty$$

$$\lim_{x \rightarrow 4^+} \frac{2}{x-4} = \frac{2}{0^+} = +\infty$$

$$\lim_{x \rightarrow 4^-} \frac{2}{x-4} = \frac{2}{0^-} = -\infty$$

$$\therefore \lim_{x \rightarrow 4^+} \frac{2}{x-4} \neq \lim_{x \rightarrow 4^-} \frac{2}{x-4}$$

$$\therefore \lim_{x \rightarrow 4} \frac{2}{x-4} \text{ DNE}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 5} \frac{-1}{x-5}$$

$$\frac{-1}{0} = \infty$$

$$\lim_{x \rightarrow 5^+} \frac{-1}{x-5} = \frac{-1}{0^+} = -\infty$$

$$\lim_{x \rightarrow 5^-} \frac{-1}{x-5} = \frac{-1}{0^-} = +\infty$$

$$\therefore \lim_{x \rightarrow 5^+} \frac{-1}{x-5} \neq \lim_{x \rightarrow 5^-} \frac{-1}{x-5}$$

$$\therefore \lim_{x \rightarrow 5} \frac{-1}{x-5} \text{ DNE}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow e} \frac{1}{\ln(x) - 1}$$

$$\frac{1}{\ln e - 1} \quad \frac{1}{1-1} \quad \frac{1}{0} \quad \infty$$

$$\lim_{x \rightarrow e^+} \frac{1}{\ln x - 1} = \frac{1}{\ln e - 1} = \frac{1}{1-1} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow e^-} \frac{1}{\ln x - 1} = \frac{1}{\ln e - 1} = \frac{1}{1-1} = \frac{1}{0^-} = -\infty$$

$$\therefore \lim_{x \rightarrow e^+} \frac{1}{\ln x - 1} \neq \lim_{x \rightarrow e^-} \frac{1}{\ln x - 1}$$

$$\therefore \lim_{x \rightarrow e} \frac{1}{\ln x - 1} \text{ DNE}$$



Text book question



Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \frac{2-x}{(x-1)^2}$$

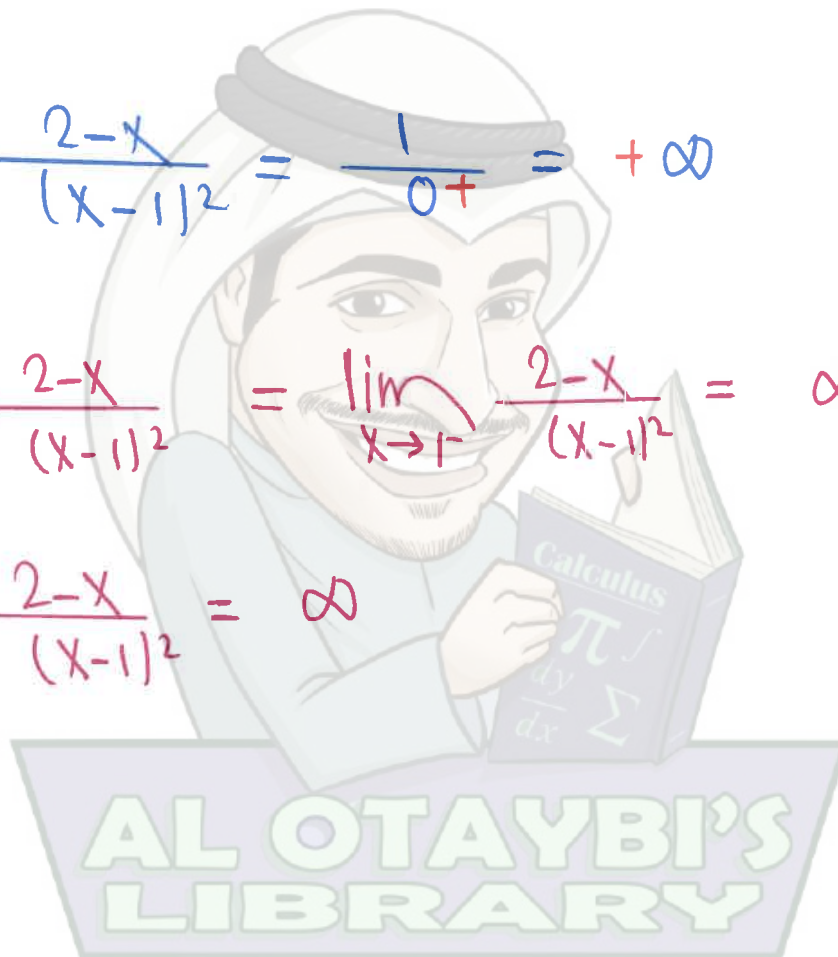
$$= \frac{1}{0} = \infty$$

$$\lim_{x \rightarrow 1^+} \frac{2-x}{(x-1)^2} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 1^-} \frac{2-x}{(x-1)^2} = \frac{1}{0^+} = +\infty$$

$$\therefore \lim_{x \rightarrow 1^+} \frac{2-x}{(x-1)^2} = \lim_{x \rightarrow 1^-} \frac{2-x}{(x-1)^2} = \infty$$

$$\therefore \lim_{x \rightarrow 1} \frac{2-x}{(x-1)^2} = \infty$$



Evaluate the limit if it exist

May 2015

$$\lim_{x \rightarrow 0} \frac{xe^x - x^2}{x \sin^2 x}$$

$$\frac{0}{0} \rightarrow f(x)$$

$$\lim_{x \rightarrow 0} \frac{\cancel{x}(e^x - x)}{\cancel{x} \sin^2 x} = \frac{1-0}{0} = \infty$$

$$\lim_{x \rightarrow 0^+} \frac{e^x - x}{\sin^2 x} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow 0^-} \frac{e^x - x}{\sin^2 x} = \frac{1}{0^+} = +\infty$$

$$\therefore \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = \infty$$

$$\therefore \lim_{x \rightarrow 0} f(x) = \infty$$

$$\text{if } f(x) = \begin{cases} \frac{x-2}{4-x^2}, & x < 2 \\ \frac{\sqrt{x+2}-2}{x-2}, & x > 2 \end{cases}$$

Evaluate the following limit if it exist

a) $\lim_{x \rightarrow 2} f(x)$

b) $\lim_{x \rightarrow 1} f(x)$

(a) $\lim_{x \rightarrow 2^+} \frac{\sqrt{x+2}-2}{x-2} \quad \frac{0}{0} \quad * \quad \frac{\sqrt{x+2}+2}{\sqrt{x+2}+2}$

$$= \lim_{x \rightarrow 2^+} \frac{x+2-4}{(x-2)(\sqrt{x+2}+2)}$$

$$= \lim_{x \rightarrow 2^+} \frac{x-2}{(x-2)(\sqrt{x+2}+2)} = \frac{1}{\sqrt{4}+2} = \frac{1}{2+2} = \left(\frac{1}{4}\right)$$

$\lim_{x \rightarrow 2^-} \frac{x-2}{4-x^2} \quad \frac{0}{0}$

$$= \lim_{x \rightarrow 2^-} \frac{x-2}{(2-x)(2+x)} = \lim_{x \rightarrow 2^-} \frac{-(2-x)}{(2-x)(2+x)} = \left(\frac{-1}{4}\right)$$

$\therefore \lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x) \implies \therefore \lim_{x \rightarrow 2} f(x) \text{ DNE}$

b) $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x-2}{4-x^2} = \left(\frac{-1}{3}\right)$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow 5^+} \frac{x+1}{x-5}$$

$$= \frac{6}{0} = \infty$$

$$= \lim_{x \rightarrow 5^+} \frac{x+1}{x-5} = \frac{6}{0^+} = +\infty$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 5^-} \frac{x+1}{x-5}$$

$$= \lim_{x \rightarrow 5^-} \frac{x+1}{x-5} = \frac{6}{0^-} = -\infty$$



Text book questions



Evaluate the limit if it exist

$$\lim_{x \rightarrow 2^-} \frac{x^2 - 2x}{x^2 - 4x + 4}$$

$$\frac{4-4}{4-8+4} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 2^-} \frac{x \cancel{(x-2)}}{(x-2) \cancel{(x-2)}}$$

$$= \lim_{x \rightarrow 2^-} \frac{x}{x-2} = \frac{2}{0^-} = -\infty$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 2^+} \frac{x^2 - 2x - 8}{x^2 - 5x + 6}$$

$$\frac{4-4-8}{4-10+6} = \frac{-8}{0}$$

$$\lim_{x \rightarrow 2^+} \frac{\underset{+}{(x+2)} \underset{-}{(x-4)}}{\underset{+}{(x-2)} \underset{-}{(x-3)}} = \frac{4(-2)}{0^-} = +\infty$$



Text book questions



$$\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{-x} \right)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{x} \right)$$

$$= \lim_{x \rightarrow 0^-} \left(\frac{2}{x} \right) = \frac{2}{0^-} = -\infty$$

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right)$$

$$= \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{x} \right)$$

$$= \lim_{x \rightarrow 0^+} 0 = 0$$

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Very important value on limit



$$e^0 = 1, \quad e^{-\infty} = 0, \quad e^{\infty} = \infty$$

The same

$$5^0 = 1, \quad 5^{-\infty} = 0, \quad 5^{\infty} = \infty$$

$$\ln 1 = 0, \quad \ln(0) = -\infty, \quad \ln(\infty) = \infty$$

$$\tan^{-1}(-\infty) = -\frac{\pi}{2}, \quad \tan^{-1}(\infty) = \frac{\pi}{2}$$

Evaluate the limit if it exist if it exist

$$\lim_{x \rightarrow 0^-} e^{\frac{1}{x}}$$

$$= e^{\frac{1}{0^-}} = e^{-\infty} = 0$$

$$\lim_{x \rightarrow 2^-} \tan^{-1}\left(\frac{1}{x-2}\right)$$

$$= \tan^{-1}\left(\frac{1}{0^-}\right) = \tan^{-1}(-\infty) = -\frac{\pi}{2}$$

$$\lim_{x \rightarrow 0^+} \tan^{-1}(\ln x)$$

$$= \tan^{-1}(\ln 0) = \tan^{-1}(-\infty) = -\frac{\pi}{2}$$



Vertical Asymptote (V.A)



المحاذي الرأسى



خطوات الحل



1) استخراج أصفار المقام.

2) أخذ اللمت لكل صفر بالاتجاهين.

يجب أن يكون ناتج اللمت

$\pm \infty$



صفر المقام = x

is Vertical Asymptote

Find the vertical asymptote (if any) of the function

$$f(x) = \frac{x-1}{x^2-1}$$

$$x^2 - 1 = 0 \longrightarrow x^2 = 1 \begin{cases} x = -1 \\ x = 1 \end{cases}$$

at $x = -1$

$$\lim_{x \rightarrow -1^+} \frac{x-1}{x^2-1}$$

$$\lim_{x \rightarrow -1^+} \frac{\cancel{x-1}}{\cancel{(x-1)}(x+1)}$$

$$\lim_{x \rightarrow -1^+} \frac{1}{x+1} = \frac{1}{0^+} = +\infty$$

$$\lim_{x \rightarrow -1^-} \frac{x-1}{x^2-1}$$

$$\lim_{x \rightarrow -1^-} \frac{\cancel{x-1}}{\cancel{(x-1)}(x+1)}$$

$$\lim_{x \rightarrow -1^-} \frac{1}{x+1} = \frac{1}{0^-} = -\infty$$

$\Rightarrow \therefore x = -1$ is Vertical Asymptote.

at $x = 1$

$$\lim_{x \rightarrow 1^+} \frac{x-1}{x^2-1}$$

$$\lim_{x \rightarrow 1^+} \frac{\cancel{x-1}}{\cancel{(x-1)}(x+1)} = \frac{1}{2}$$

$$\lim_{x \rightarrow 1^-} \frac{x-1}{x^2-1}$$

$$\lim_{x \rightarrow 1^-} \frac{\cancel{x-1}}{\cancel{(x-1)}(x+1)} = \frac{1}{2}$$

$\Rightarrow \therefore x = 1$ is Not Vertical Asymptote.

Find the vertical asymptote (if any) of the function

$$f(x) = \frac{1}{1 - e^x}$$

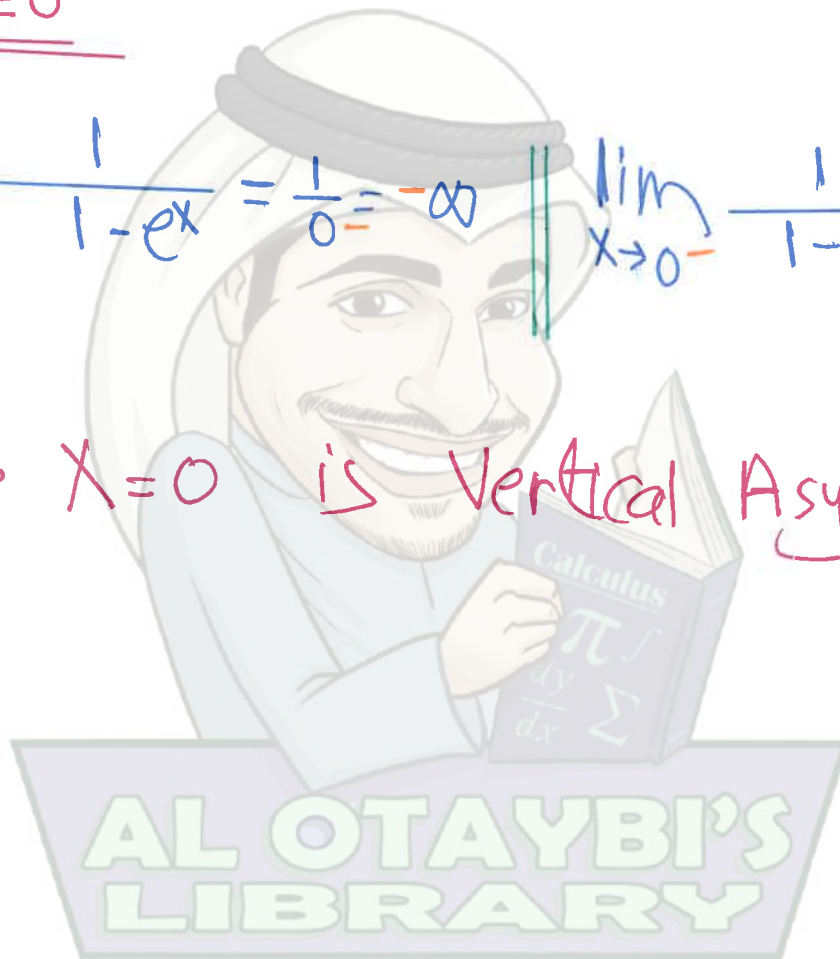
$$1 - e^x = 0 \longrightarrow 1 = e^x$$

$$\ln 1 = \ln e^x \longrightarrow \boxed{x=0}$$

at $x=0$

$$\lim_{x \rightarrow 0^+} \frac{1}{1 - e^x} = \frac{1}{0^-} = -\infty \quad \parallel \quad \lim_{x \rightarrow 0^-} \frac{1}{1 - e^x} = \frac{1}{0^+} = +\infty$$

$\Rightarrow \therefore x=0$ is Vertical Asymptote.





Text book question



Find the vertical asymptotes of the function

$$y = \frac{x^2 + 1}{3x - 2x^2}$$

$$3x - 2x^2 = 0$$

$$x(3 - 2x) = 0$$

$$x = 0$$

$$3 - 2x = 0 \longrightarrow$$

$$x = \frac{3}{2}$$

$$x = 0$$

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{x^2 + 1}{3x - 2x^2} \\ = \lim_{x \rightarrow 0^+} \frac{x^2 + 1}{x(3 - 2x)} = \frac{1}{0^+} = +\infty \end{aligned}$$

$$\lim_{x \rightarrow 0^-} \frac{x^2 + 1}{3x - 2x^2}$$

$$= \lim_{x \rightarrow 0^-} \frac{x^2 + 1}{x(3 - 2x)} = \frac{1}{0^-} = -\infty$$

$\Rightarrow \therefore x = 0$ is Vertical Asymptote

$$x = \frac{3}{2}$$

$$\lim_{x \rightarrow \frac{3}{2}^+} \frac{x^2 + 1}{3x - 2x^2}$$

$$= \lim_{x \rightarrow \frac{3}{2}^+} \frac{x^2 + 1}{x(3 - 2x)} = \frac{(\frac{3}{2})^2 + 1}{0^-} = -\infty$$

$$\lim_{x \rightarrow \frac{3}{2}^-} \frac{x^2 + 1}{3x - 2x^2}$$

$$= \lim_{x \rightarrow \frac{3}{2}^-} \frac{x^2 + 1}{x(3 - 2x)} = \frac{(\frac{3}{2})^2 + 1}{0^+} = +\infty$$

$\Rightarrow \therefore x = \frac{3}{2}$ is Vertical Asymptote.

Find the vertical asymptote for

$$f(x) = \frac{\sqrt{x^4 + 5x^2}}{x^2 - x}$$

$$x^2 - x = 0$$

$$x(x-1) = 0$$

$$x = 0$$

$$x-1 = 0$$

$$x = 1$$

at $x=0$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x^4 + 5x^2}}{x^2 - x} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x^2(x^2 + 5)}}{x(x-1)}$$

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{x^2} \sqrt{x^2 + 5}}{x(x-1)}$$

$$\lim_{x \rightarrow 0^+} \frac{|x| \sqrt{x^2 + 5}}{x(x-1)}$$

$$\lim_{x \rightarrow 0^+} \frac{\cancel{x} \sqrt{x^2 + 5}}{\cancel{x}(x-1)}$$

$$= \frac{\sqrt{5}}{-1} = -\sqrt{5}$$

$$\lim_{x \rightarrow 0^-} \frac{\sqrt{x^4 + 5x^2}}{x^2 - x} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0^-} \frac{\sqrt{x^2(x^2 + 5)}}{x(x-1)}$$

$$\lim_{x \rightarrow 0^-} \frac{\sqrt{x^2} \sqrt{x^2 + 5}}{x(x-1)}$$

$$\lim_{x \rightarrow 0^-} \frac{|x| \sqrt{x^2 + 5}}{x(x-1)}$$

$$\lim_{x \rightarrow 0^-} \frac{\cancel{-x} \sqrt{x^2 + 5}}{\cancel{-x}(x-1)}$$

$$= \frac{-\sqrt{5}}{-1} = \sqrt{5}$$

$\therefore x=0$ is Not Vertical asymptote

Eng. Mohammad Alotaybi 90048657 Twitter: @M_Alotaybi

→ See Next Page Plz!

at $x=1$

$$\lim_{x \rightarrow 1^+} \frac{\sqrt{x^4 + 5x^2}}{x^2 - x} \quad \frac{\sqrt{1+5}}{0}$$

$$\lim_{x \rightarrow 1^+} \frac{\sqrt{x^4 + 5x^2}}{x(x-1)}$$

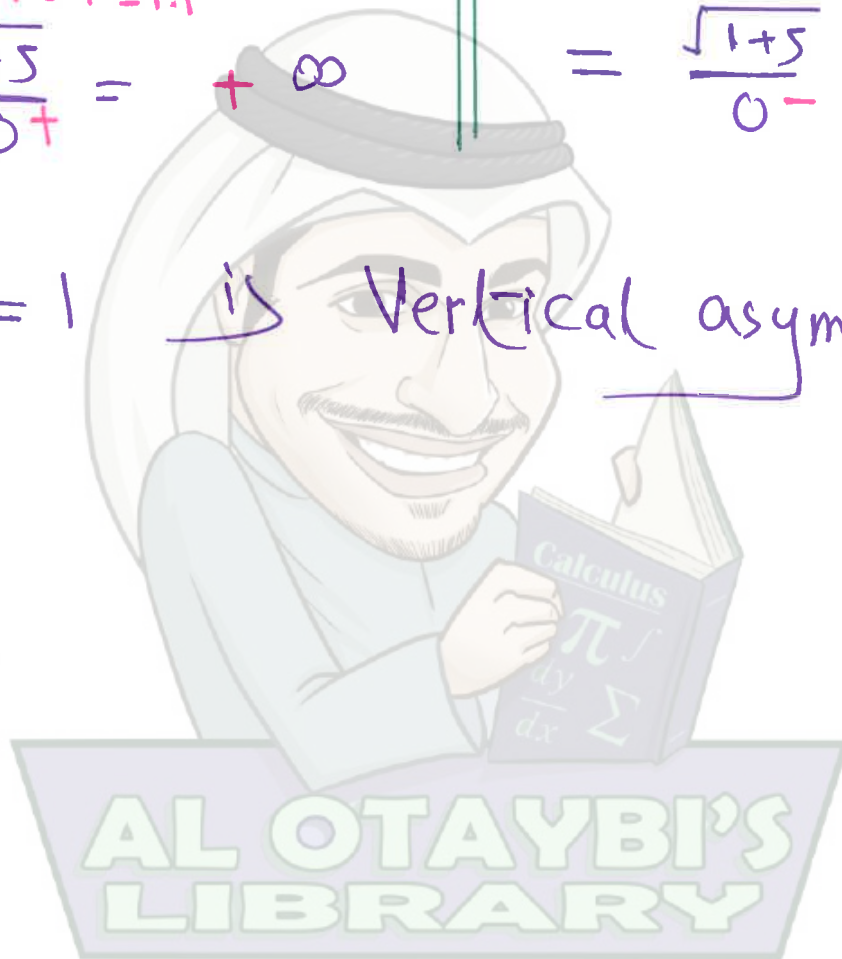
$$= \frac{\sqrt{1+5}}{0^+} = +\infty$$

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^4 + 5x^2}}{x^2 - x}$$

$$\lim_{x \rightarrow 1^-} \frac{\sqrt{x^4 + 5x^2}}{x(x-1)}$$

$$= \frac{\sqrt{1+5}}{0^-} = -\infty$$

$\therefore x=1$ is vertical asymptote





كيفية قراءة اللمت والصورة من الرسمة



(1) أروح حق الـ **x-axis** بمقدار

(2) (أصعد-أنزل-أثبت) على الدالة

(3) من مكان التقاطع , أمد خط أفقي يقطع الـ **y-axis**



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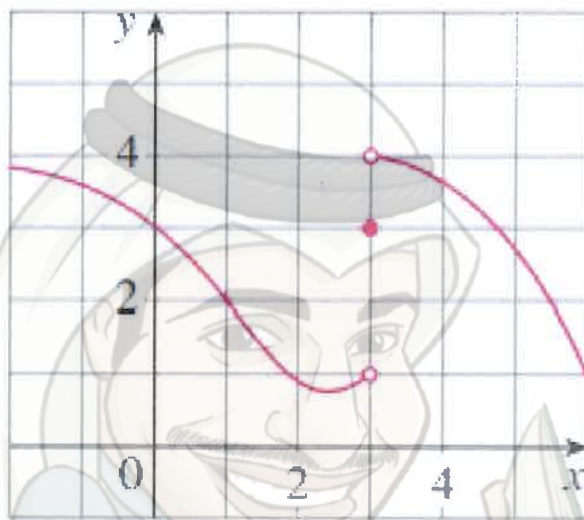


Text book question



For the function f whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

- (a) $\lim_{x \rightarrow 1} f(x)$ (b) $\lim_{x \rightarrow 3^-} f(x)$ (c) $\lim_{x \rightarrow 3^+} f(x)$
 (d) $\lim_{x \rightarrow 3} f(x)$ (e) $f(3)$



(a) $\lim_{x \rightarrow 1} f(x) = 2$

(b) $\lim_{x \rightarrow 3^-} f(x) = 1$

(c) $\lim_{x \rightarrow 3^+} f(x) = 4$

(d) $\lim_{x \rightarrow 3} f(x) \text{ DNE}$

(e) $f(3) = 3$

$\lim_{x \rightarrow 3^+} f(x) \neq \lim_{x \rightarrow 3^-} f(x)$
 $4 \neq 1$

يفرق الاتجاهين : 3
يفرق الصورة :

Extra

$\lim_{x \rightarrow 2^+} f(x) = 1$

$\lim_{x \rightarrow 2^-} f(x) = 1$

$\lim_{x \rightarrow 2} f(x) = 1$

$f(2) = 1$

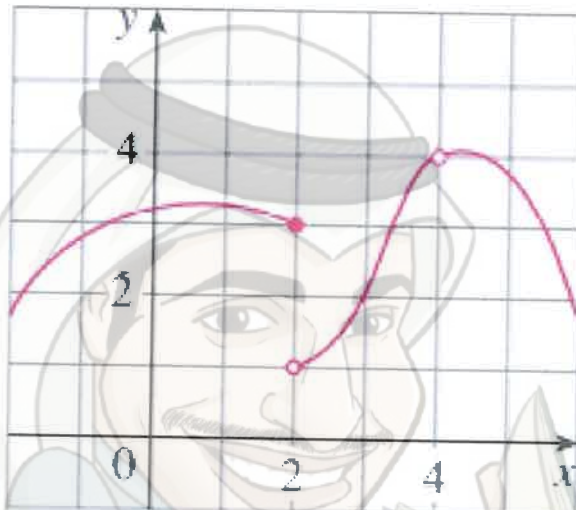


Text book question



Use the given graph of f to state the value of each quantity, if it exists. If it does not exist, explain why.

- (a) $\lim_{x \rightarrow 2^-} f(x)$ (b) $\lim_{x \rightarrow 2^+} f(x)$ (c) $\lim_{x \rightarrow 2} f(x)$
 (d) $f(2)$ (e) $\lim_{x \rightarrow 4} f(x)$ (f) $f(4)$



(a) $\lim_{x \rightarrow 2^-} f(x) = 3$

(b) $\lim_{x \rightarrow 2^+} f(x) = 1$

(c) $\lim_{x \rightarrow 2} f(x)$ DNE ($\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$
 $1 \neq 3$)

(d) $f(2) = 3$

(e) $\lim_{x \rightarrow 4} f(x) = 4$

(f) $f(4)$ undefined

يفرق الـ $\lim_{x \rightarrow 2^-}$: 3
 لـ $\lim_{x \rightarrow 2^+}$: 1

Extra

$\lim_{x \rightarrow 4^+} f(x) =$

$\lim_{x \rightarrow 4^-} f(x) =$

$\lim_{x \rightarrow 4} f(x) =$



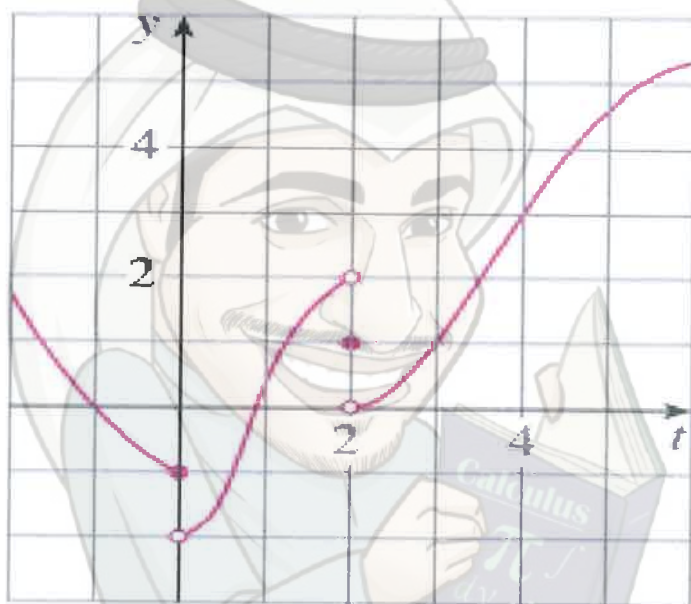


Text book question



For the function g whose graph is given, state the value of each quantity, if it exists. If it does not exist, explain why.

- (a) $\lim_{t \rightarrow 0^-} g(t)$ (b) $\lim_{t \rightarrow 0^+} g(t)$ (c) $\lim_{t \rightarrow 0} g(t)$
 (d) $\lim_{t \rightarrow 2^-} g(t)$ (e) $\lim_{t \rightarrow 2^+} g(t)$ (f) $\lim_{t \rightarrow 2} g(t)$
 (g) $g(2)$ (h) $\lim_{t \rightarrow 4} g(t)$



$$(a) \lim_{t \rightarrow 0^-} g(t) = -1$$

$$(b) \lim_{t \rightarrow 0^+} g(t) = -2$$

$$(c) \lim_{t \rightarrow 0} g(t) \text{ DNE}$$

($\lim_{t \rightarrow 0^+} g(t) \neq \lim_{t \rightarrow 0^-} g(t)$)
 $-1 \neq -2$

$$(d) \lim_{t \rightarrow 2^-} g(t) = 2$$

$$(e) \lim_{t \rightarrow 2^+} g(t) = 0$$

يفرق الـ تجاوزين : 0, 2
يفرق الصورة :

$$(f) \lim_{t \rightarrow 2} g(t) \text{ DNE}$$

$$(\lim_{t \rightarrow 2^+} g(t) \neq \lim_{t \rightarrow 2^-} g(t))$$

$$0 \neq 2$$

$$(g) g(2) = 1$$

$$(h) \lim_{t \rightarrow 4} g(t) = 3$$



Text book question

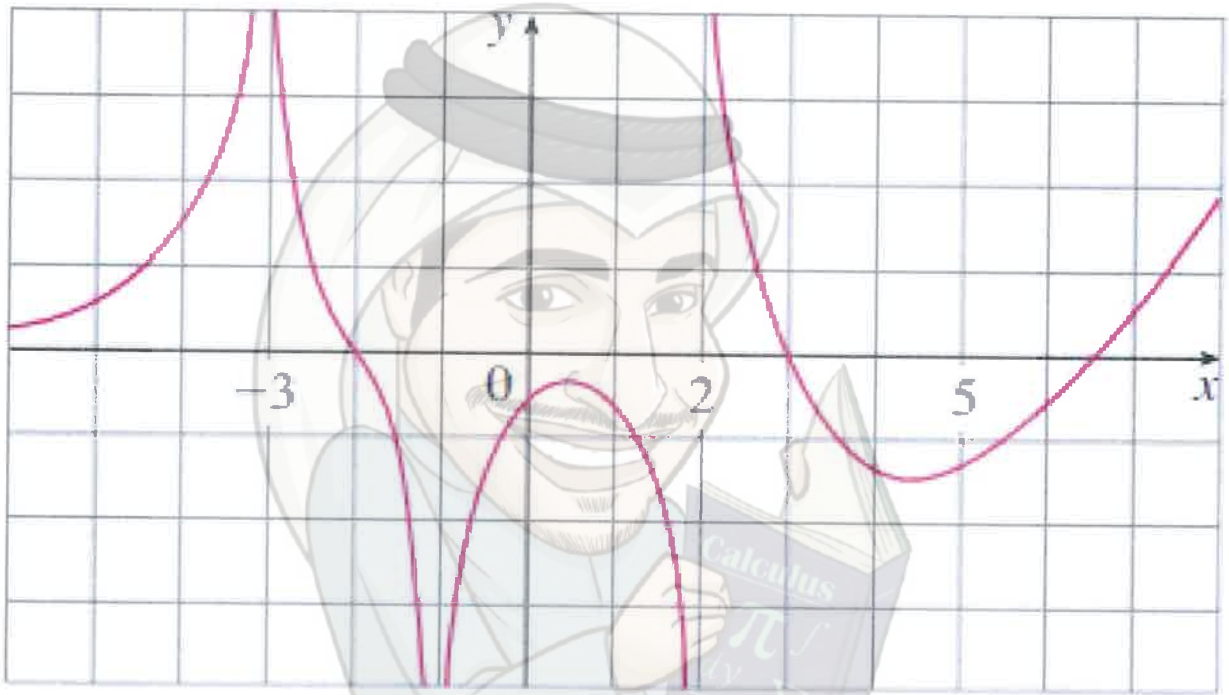


For the function A whose graph is shown, state the following.

(a) $\lim_{x \rightarrow -3} A(x)$ (b) $\lim_{x \rightarrow 2^-} A(x)$

(c) $\lim_{x \rightarrow 2^+} A(x)$ (d) $\lim_{x \rightarrow -1} A(x)$

(e) The equation of the vertical asymptotes.



(a) $\lim_{x \rightarrow -3} A(x) = +\infty$

(b) $\lim_{x \rightarrow 2^-} A(x) = -\infty$

(c) $\lim_{x \rightarrow 2^+} A(x) = +\infty$

(d) $\lim_{x \rightarrow -1} A(x) = -\infty$

(e)

The equation of Vertical Asy.

$x = -3$ ($\lim_{x \rightarrow -3^+} A(x) = \infty, \lim_{x \rightarrow -3^-} A(x) = \infty$)

$x = -1$ ($\lim_{x \rightarrow -1^+} A(x) = -\infty, \lim_{x \rightarrow -1^-} A(x) = -\infty$)

$x = 2$ ($\lim_{x \rightarrow 2^+} A(x) = \infty, \lim_{x \rightarrow 2^-} A(x) = -\infty$)

يفرق الاتجاهين : -3, -1, 2

يفرق الصورة :



Text book question

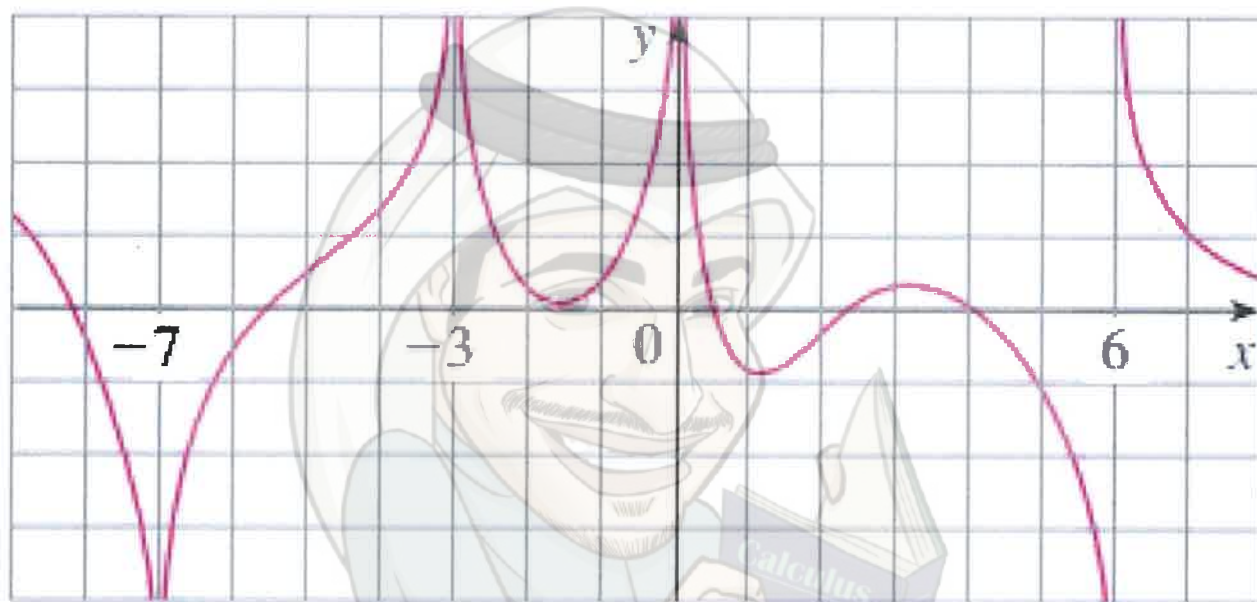


For the function f whose graph is shown, state the following.

(a) $\lim_{x \rightarrow -7} f(x)$ (b) $\lim_{x \rightarrow -3} f(x)$ (c) $\lim_{x \rightarrow 0} f(x)$

(d) $\lim_{x \rightarrow 6^-} f(x)$ (e) $\lim_{x \rightarrow 6^+} f(x)$

(f) The equations of the vertical asymptotes



(a) $\lim_{x \rightarrow -7} f(x) = -\infty$

(b) $\lim_{x \rightarrow -3} f(x) = \infty$

(c) $\lim_{x \rightarrow 0} f(x) = \infty$

(d) $\lim_{x \rightarrow 6^-} f(x) = -\infty$

(e) $\lim_{x \rightarrow 6^+} f(x) = \infty$

يفرق الاتجاهين : $-7, -3, 0, 6$
 لفرق الصورة :

(f) The equation of the vertical asy.

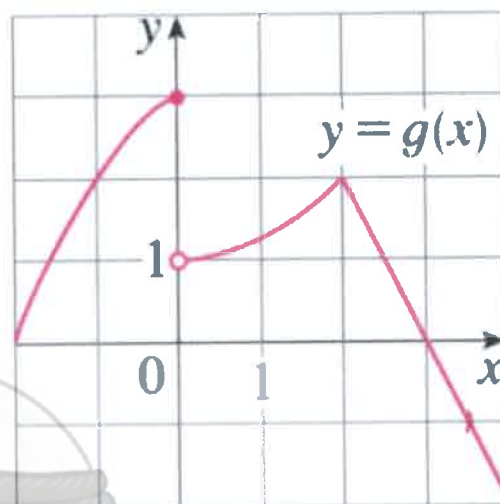
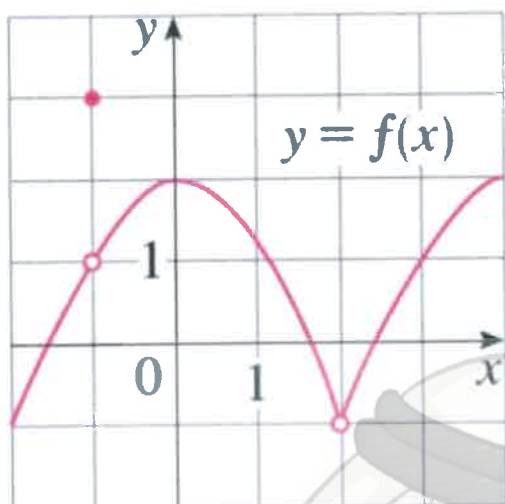
$x = -7$ ($\lim_{x \rightarrow -7^+} f(x) = -\infty$, $\lim_{x \rightarrow -7^-} f(x) = -\infty$)

$x = -3$ ($\lim_{x \rightarrow -3^+} f(x) = \infty$, $\lim_{x \rightarrow -3^-} f(x) = \infty$)

$x = 0$ ($\lim_{x \rightarrow 0^+} f(x) = \infty$, $\lim_{x \rightarrow 0^-} f(x) = \infty$)

$x = 6$ ($\lim_{x \rightarrow 6^+} f(x) = -\infty$, $\lim_{x \rightarrow 6^-} f(x) = -\infty$)

Use the given graphs of f and g to find each of the following limits, if it exist.



(a) $\lim_{x \rightarrow 2} [f(x) + g(x)]$

(b) $\lim_{x \rightarrow 3} \frac{g(x)}{f(x)}$

(c) $\lim_{x \rightarrow 0} [f(x)g(x)]$

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(a) $\lim_{x \rightarrow 2} [f(x) + g(x)] = -1 + 2 = 1$

f
يفرق الاتجاهين:
يفرق الصورة : 2, -1

(b) $\lim_{x \rightarrow 3} \frac{g(x)}{f(x)} = \frac{0}{1} = 0$

g
يفرق الاتجاهين:
يفرق الصورة:

(c) $\lim_{x \rightarrow 0} [f(x)g(x)]$ DNE

$(\lim_{x \rightarrow 0^+} f(x)g(x) \neq \lim_{x \rightarrow 0^-} f(x)g(x))$
 $2(1) \neq 2(3)$
 $2 \neq 6$



Text book question



Is there a number a such that

$$\lim_{x \rightarrow -2} \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$$

exist? If so, find the value of a and the value of the limit

$$\lim_{x \rightarrow -2} \frac{3x^2 + ax + a + 3}{x^2 + x - 2} \quad \frac{3(4) - 2a + a + 3}{4 - 2 - 2} \quad \frac{12 - a + 3}{0} \quad \frac{15 - a}{0}$$

The limit exist only when $15 - a = 0$

$$a = 15$$

$$\Rightarrow \lim_{x \rightarrow -2} \frac{3x^2 + 15x + 15 + 3}{x^2 + x - 2}$$

$$\lim_{x \rightarrow -2} \frac{3x^2 + 15x + 18}{x^2 + x - 2} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow -2} \frac{3(x^2 + 5x + 6)}{x^2 + x - 2}$$

$$\lim_{x \rightarrow -2} \frac{3(\cancel{x+2})(x+3)}{(x-1)(\cancel{x+2})}$$

$$= \frac{3(1)}{1-3} = -1$$



Text book question



Explain in your own words what is meant by the equation

$$\lim_{x \rightarrow 2} f(x) = 5$$

When x approaches to 2

the value of $f(x)$ approach to 5



Homework



Evaluate the limit if it exists

$$\lim_{x \rightarrow -4} \frac{\sqrt{x^2 + 9} - 5}{x + 4}$$

(Answer : $-\frac{4}{5}$)

Evaluate the limit if it exists

$$\lim_{h \rightarrow 0} \frac{(-5 + h)^2 - 25}{h}$$

(Answer : -10)